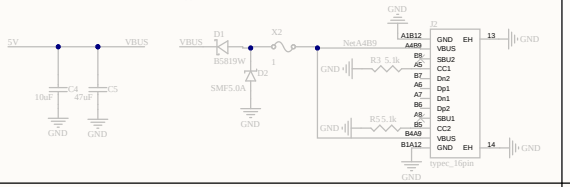
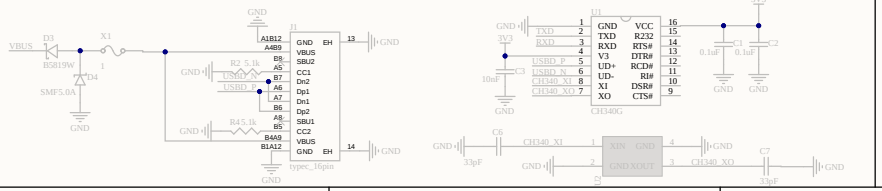
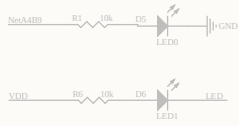
USB仅供电



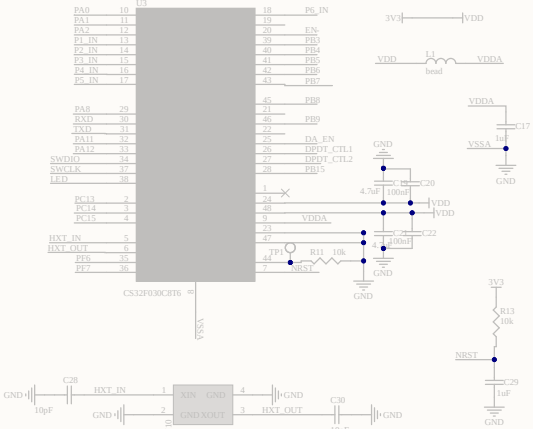
USB转串口



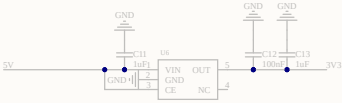
电源指示灯



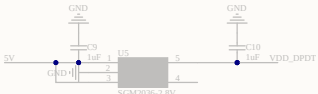
控制核心



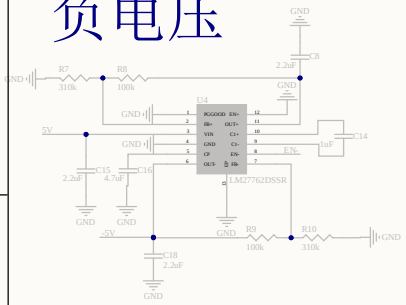
5v转3.3v



5v转2.8v



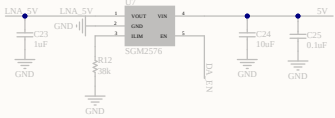
负电压



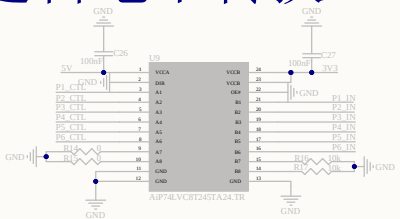
SWD下载口



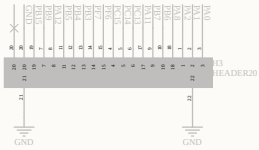
DA使能开关



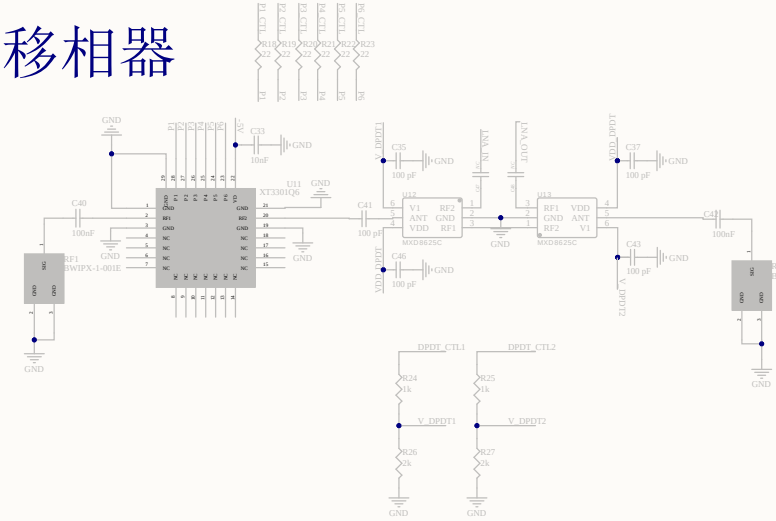
逻辑电平转换



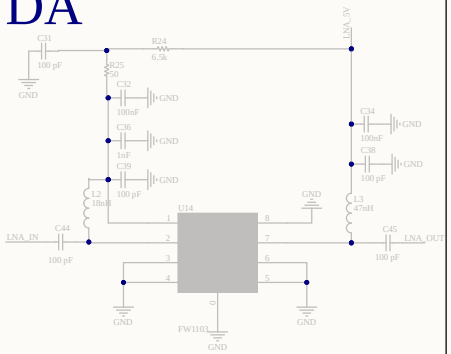
IO口



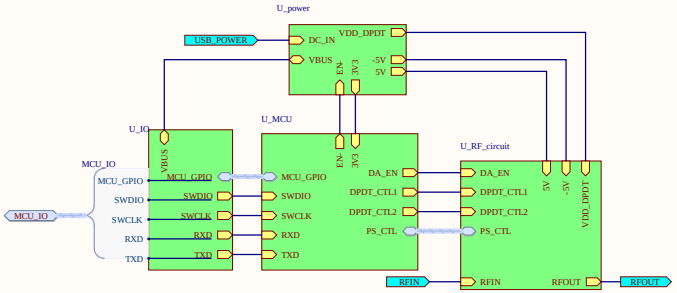
移相器



DA



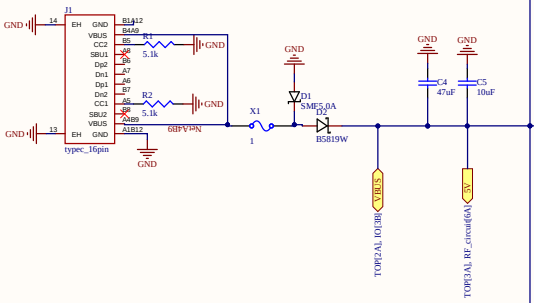
Title		
Size: A2	Number	Revision
Date: 10/10/2026	Sheet of	
File: C:\Users\...\SHIFTER.SchDoc	Drawn By:	



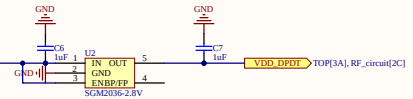
Title			
Size: A2	Number		Revision
Date: 10/10/2026		Sheet of	
File: C:\Users\...\TOP.SchDoc		Drawn By:	

5v转3.3v

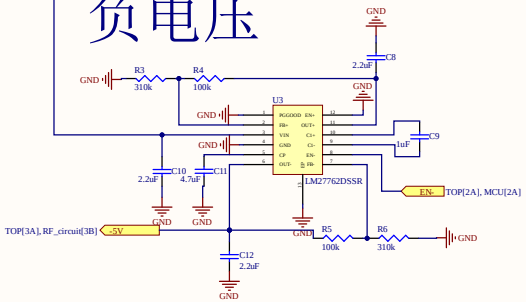
USB仅供电



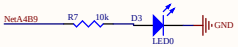
5v转2.8v



负电压

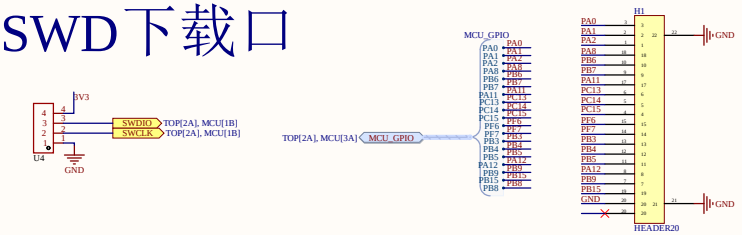


电源指示灯

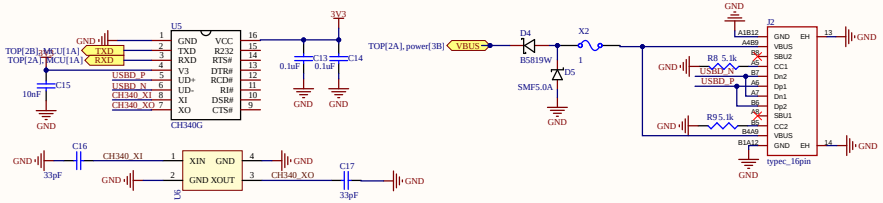


Title		
Size: A2	Number	Revision
Date: 10/10/2026		Sheet of
File: C:\Users\...\power.SchDoc		Drawn By:

SWD下载口

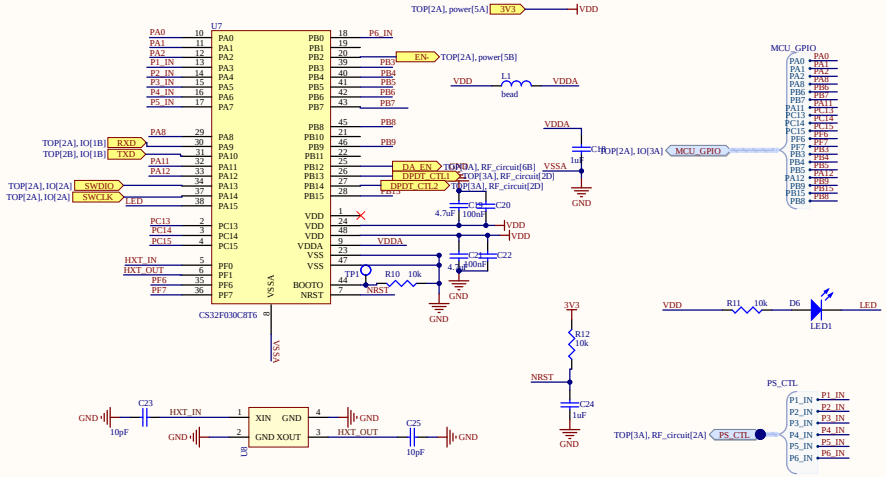


USB转串口



Title			
Size	Number		Revision
A2			
Date:	10/10/2026	Sheet of	
File:	C:\Users\...\IO.SchDoc	Drawn By:	

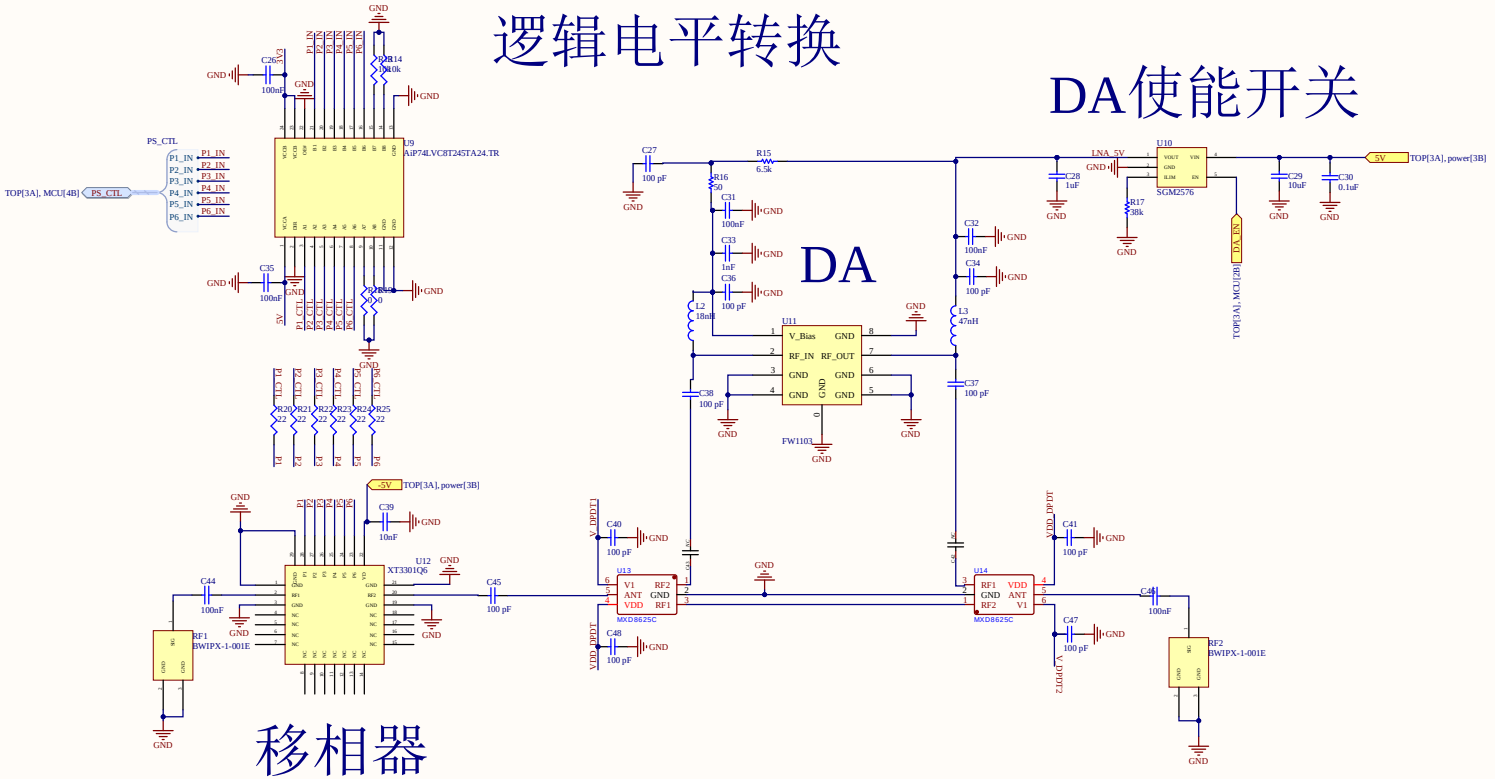
控制核心



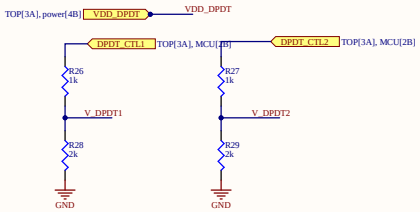
Title			
Size	Number		Revision
A2			
Date:	10/10/2026	Sheet of	
File:	C:\Users\...\MCU.SchDoc	Drawn By:	

逻辑电平转换

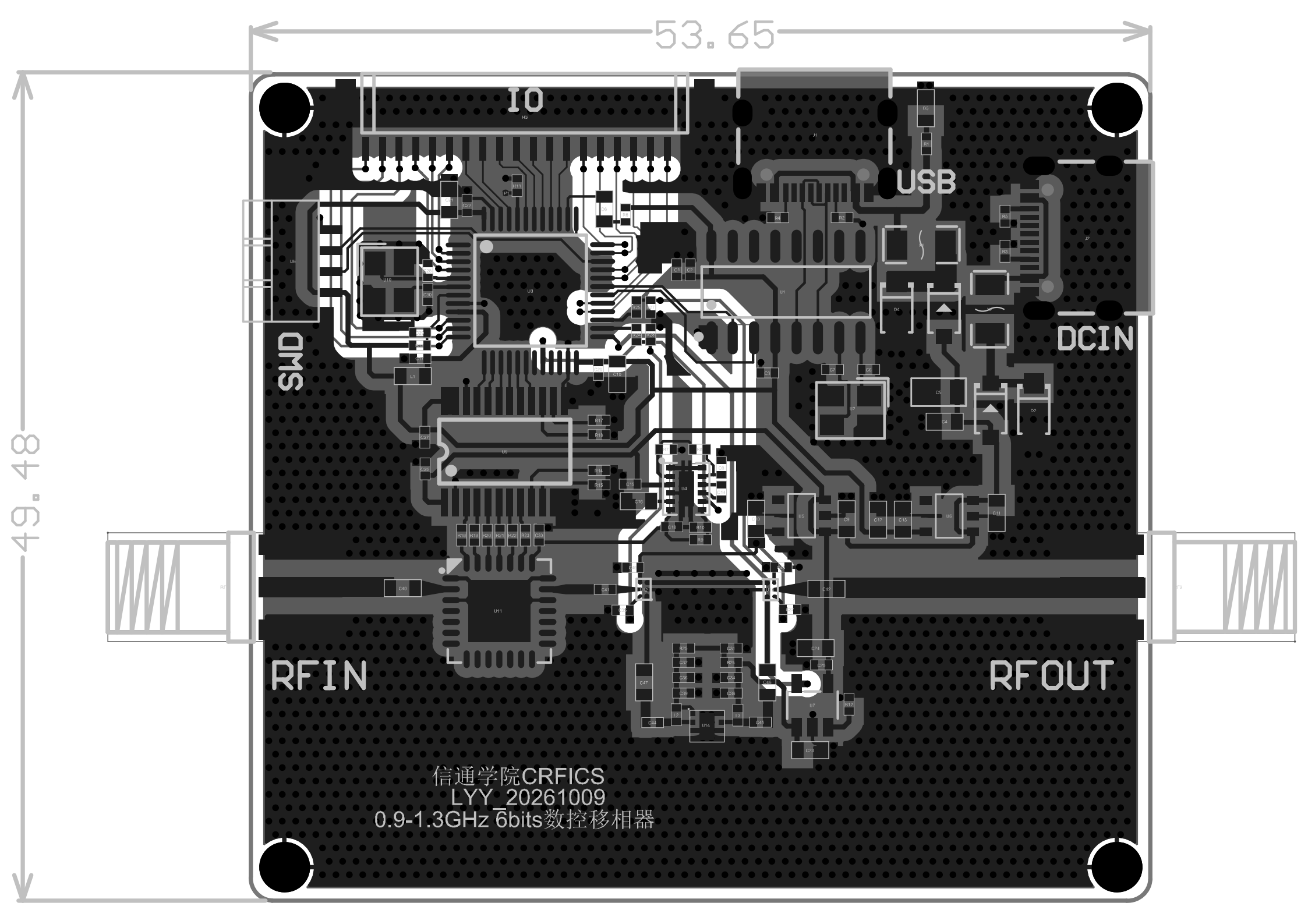
DA使能开关



移相器



Title		
Size: A2	Number	Revision
Date: 10/10/2026		Sheet of
File: C:\Users\...\RF_circuit.SchDoc		Drawn By:


53.65
49.48
IO
USB
DCIN
SWD
RFIN
RFOUT
信通学院CRFICS
LYY 20261009
0.9-1.3GHz 6bits数控移相器

Comment	Description	Designator	Footprint	LibRef	Quantity
1uF	Capacitor	C1, C3, C6, C7, C28	0603_Z	Cap	5
100nF	Capacitor, 0402	C2, C44, C46	0603_Z	Cap	3
47uF	0402	C4	0805	CAP	1
10uF	0402, Capacitor	C5, C29	0603_Z	CAP	2
2.2uF	0402	C8, C10, C12	0402_Z	CAP	3
1uF	0402	C9, C18, C24	0402_Z	CAP	3
4.7uF	0402	C11, C19, C21	0603_Z	CAP	3
0.1uF	0402, Capacitor	C13, C14, C30	0402_Z	CAP	3
10nF	0402	C15, C39	0402_Z	CAP	2
33pF	0402	C16, C17	0402_Z	CAP	2
100nF	0402, Capacitor	C20, C22, C26, C31, C32, C35	0402_Z	CAP	6
10pF	0402	C23, C25	0402_Z	CAP	2
100 pF	Capacitor	C27, C34, C36, C37, C38, C40, C41, C45, C47, C48	0402_Z	Cap	10
1nF	Capacitor	C33	0402_Z	Cap	1
SMF5.0A		D1, D5	SMF5.0A	SMF5.0A	2
B5819W	肖特基二极管	D2, D4	BW5819W	B5819W	2
LED0	Typical INFRARED GaAs LED	D3	0603_Z	LED0	1
LED1	Typical INFRARED GaAs LED	D6	0603_Z	LED0	1
HEADER20		H1	Header20	HEADER20	1
typec_16pin		J1, J2	TYPEC_16PIN	typec_16pin	2
bead	INDUCTOR	L1	0603_Z	INDUCTOR1_1	1
18nH	Inductor	L2	0402_Z	Inductor	1
47nH	Inductor	L3	0402_Z	Inductor	1
5.1k	Resistor	R1, R2, R8, R9	0402_Z	Res3	4
310k	Resistor	R3, R6	0402_Z	Res3	2
100k	Resistor	R4, R5	0402_Z	Res3	2
10k	Resistor	R7, R10, R11, R12, R13, R14	0402_Z	Res3	6
6.5k	Resistor	R15	0402_Z	Res1	1
50	Resistor	R16	0402_Z	Res1	1
38k	Resistor	R17	0402_Z	Res1	1
0	Resistor	R18, R19	0402_Z	Res3	2
22	Resistor	R20, R21, R22, R23, R24, R25	0402_Z	Res3	6
1k	Resistor	R26, R27	0402_Z	Res3	2
2k	Resistor	R28, R29	0402_Z	Res3	2
BWIPX-1-001E		RF1, RF2	SMA-SMD_BWSMA-KE-P001	BWIPX-1-001E	2
TP_P		TP1	TP_0.8	TP_P	1
SGM2036-3.3YN5G/TR	0.8V-5V可调输出线性稳压器	U1	SOT-23-5	LDO_ADJ_SGM2036	1
SGM2036-2.8V		U2	SOT-23-5	sgm2036	1
LM27762DSSR		U3	LM27762	LM27762DSSR	1
HDGC1251WR-4P	复旦微单片机程序下载接口	U4	J4_1.25	MX1.25_弯针	1
CH340G		U5	CH340G	CH340G	1
12MHz		U6	CRYSTAL-SMD_4P-L3.2-W2.5-BL	XTAL 3225	1
CS32F030C8T6		U7	LQPF48	CS32F030C8T6	1
8MHz		U8	CRYSTAL-SMD_4P-L3.2-W2.5-BL	XTAL 3225	1
AiP74LVC8T245TA24.TR	具有三态输出的8位双电源转换收发器，可实现双向电平转换	U9	AIP74LVC8T245TA24.TR	AiP74LVC8T245TA24.TR	1
SGM2576		U10	SGM2576	SGM2576	1
FW1103		U11	FW1103W	FW1103	1
XT3301Q6		U12	XT3301Q6	XT3301Q6	1
MXD8625C		U13, U14	MXD8625C	MXD8625C	2
Fuse	Fuse	X1, X2	F1206	Fuse	2